Random Variables

A *random variable* assigns a real number to a random outcome. Formally, it is a function $X : \Omega \to \mathbb{R}$, where Ω is the set of outcomes.

Intuitively, a random variable may be seen as an "encoding" of a random event $\omega \in \Omega$ as a real number. For example, a random variable X which "encodes" the roll of a 6-sided die with sample space $\Omega = \{$⚀, ⚁, ⚂, ⚃, ⚄, ⚅$\}$ "counts" the dots on the top face of a rolled die, e.g. $X($⚄$) = 3$.

Events can then be described using the variable's values, such as $[X = 2]$, $[X < 0]$ or $[2 \leq X \leq 5]$.

Discrete Distributions

A random variable X is *discrete* if it takes a finite or countable set of values. Its *probability mass function* (p.m.f.) gives the probability of each value x that X may take

$$f_X(x) = P(X = x).$$

and is zero otherwise. The p.m.f. determines the probabilities of events described in terms of X as

$$P(X \in A) = \sum_{x \in A} f_X(x),$$

where the sum is over all possible values x in A. Values in A that X cannot take have probability zero and do not contribute.

Example (Sum of two dice)**.** The outcomes are ordered pairs of die results, so

$$\Omega = \{(⚀, ⚀), (⚀, ⚁), \ldots, (⚅, ⚄), (⚅, ⚅)\}.$$

There are 36 equally likely outcomes. Define X to be the sum of the two results:

$$X : \Omega \to \mathbb{R}, \qquad X((a, b)) = a + b.$$

For example, $X((⚄, ⚄)) = 5$. Although Ω has 36 outcomes, X takes only the values $2, 3, \ldots, 12$.

The event that the sum is at most 4 is

$$[X \leq 4] = \{(⚀, ⚀), (⚀, ⚁), (⚀, ⚂), (⚁, ⚀), (⚁, ⚁), (⚂, ⚀)\}.$$

Therefore,

$$P(X \leq 4) = \frac{6}{36} = \frac{1}{6}.$$

The p.m.f. of X can be given by a table or by the formula

$$f_X(x) = P(X = x) = \frac{\min(x - 1, 13 - x)}{36}, \qquad x \in \{2, 3, \ldots, 12\}.$$

For instance,

$$P(X \leq 4) = \sum_{x=2}^{4} f_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} = \frac{1}{6}.$$

Once the p.m.f. is known, probabilities for X can be calculated without the underlying probability space.

Continuous Distributions

A random variable X is *continuous* if its probabilities can be described by a *probability density function* (p.d.f. or *density*) f_X. The density is nonnegative, and its total area is 1:

$$f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

For a random variable with density f_X, the probability of X being in an interval $[a, b]$ is

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx.$$

A consequence of the formulation as an integral is that endpoints inclusion do not change the probability:

$$P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b).$$

Example (Waiting for the bus)**.** Suppose a bus arrives every 10 minutes, and your arrival time is uniform between two consecutive bus arrivals. Let X be your waiting time, in minutes. Under this simplified model, X is uniformly distributed between 0 and 10, with density

$$f_X(x) = \begin{cases} \frac{1}{10}, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

The probability of waiting between 2 and 5 minutes is

$$P(2 \leq X \leq 5) = \int_2^5 \frac{1}{10}\,dx = \frac{3}{10}.$$

while the probability of waiting less than 5 minutes is

$$P(X \leq 5) = \int_{-\infty}^5 f_X(x)\,dx = \int_0^5 \frac{1}{10}\,dx = \frac{1}{2}.$$

An interesting consequence for continuous random variables is that any "infinitely precise" value has probability zero. What is the probability that you wait exactly $\pi = 3.1415\ldots$ minutes? We can express this as the limit of a $\pm\varepsilon$ interval around exactly π minutes with ε going to zero, that is

$$P(X = \pi) = \lim_{\varepsilon \to 0} P(X \in [\pi - \varepsilon, \pi + \varepsilon]) = \lim_{\varepsilon \to 0} \int_{\pi-\varepsilon}^{\pi+\varepsilon} \frac{1}{10}\,dx = \lim_{\varepsilon \to 0} \frac{\varepsilon}{5} = 0.$$

Exercise 4

Problem 1 (Marbles and cells)**.** Seven marbles are placed, one at a time, into seven cells. For each marble, every cell is equally likely to be chosen, independently of the other placements. What is the probability of:

1. *one cell* contains *four marbles*.
2. *two cells* contain either *two or three marbles*, all other cells contain neither two nor three marbles.

How does this problem differ from Problem 5 (Balls and cells) in Exercise 2?

Problem 2 (Maintenance)**.** On average, a company's web service requires $\lambda = 1.5$ maintenance shutdowns per year. Modeling the Nr. of shutdowns X per year with a Poisson distribution, that is $X \sim \text{Poisson}(\lambda)$ with p.m.f. f_X given as

$$f_X(x) = e^{-\lambda}\frac{\lambda^x}{x!}$$

for $x \in \mathbb{N}_0$. What are the probabilities for the following scenarios:

1. Exactly 2 shutdowns in a year,
2. Between 3 and 6 shutdowns (inclusive) in a year.
3. More than 5 shutdowns in a year.
4. Finally, an even nr. of shutdowns, that is $0, 2, 4, \ldots$.

Problem 3 (Transforms and Moments)**.** Let $X \sim \mathcal{N}(0,1)$ be standard normal distributed with density

$$f_X(x) = \frac{1}{\sqrt{2\pi}}\exp\Big(-\frac{x^2}{2}\Big)$$

1. Compute $\mathbb{E}\,X^2 = \int_{-\infty}^{\infty} x^2 f_X(x)\,dx$ explicitely via integration.[1]
2. Now use the p.d.f. of $Y = X^2$ given as $f_Y(y) = \frac{1}{\sqrt{2\pi y}}e^{-y/2}$ and compute $\mathbb{E}\,Y = \int_0^{\infty} y f_Y(y)\,dy$.
3. Find the p.d.f. of $|X|$ using the symmetry of the standard normal density f_X.

Problem 4 (Maximum)**.** Let X_1 and X_2 be independent and identically distributed (i.i.d.) random variables with *cumulative distribution function* (c.d.f.) $F_X(x) = P(X \le x)$.

1. Define $Y = \max(X_1, X_2)$ and find the c.d.f. F_Y of Y with respect to F_X.
2. Now assume $X_1, X_2 \overset{\text{iid}}{\sim} \text{U}(0,1)$ are continuous uniform. That is, both X_1 and X_2 have density $f(x) = 1$ if $0 \le x \le 1$ and 0 else. Compute F_Y and its density $f_Y(y) = \frac{d}{dy}F_Y(y)$ explicitely.

Problem 5 (Inverse Sampling[2])**.** Let X be a continuous random variable with values in $[-\pi, \pi]$ with density f_X, that is $X \sim f_X$, where

$$f_X(x) \propto |\sin(x)|.$$

The symbol $\propto$ stands for "proportional to", that is there exists a constant $a \in \mathbb{R}$ such that $f_X(x) = a\,|\sin(x)|$.

1. Determin the constant $a \in \mathbb{R}$.
2. Compute the cumulative distribution function (c.d.f.) F_X.
3. Let $U \sim \text{U}(0,1)$, what is the distribution of $F_X^{-1}(U)$?
4. Use the last point to write an `R` function `rsin(k)` which draws n independent and identical distributed samples from the distribution of X.
5. Create a histogram of at least 10 000 samples and add the density f_X to the plot to visualize that your function does generate samples drawn from the right distribution (see `?hist`, `curve` and `lines`.)

[1] A simpler way is to observe that $1 = \mathbb{V}\text{ar}(X) = \mathbb{E}\,X^2 - (\mathbb{E}\,X)^2 = \mathbb{E}\,X^2$.
[2] See Problem 4